

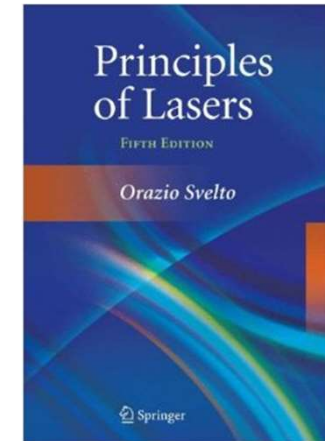


南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

《激光原理与技术》

Principles and Technologies of Lasers

Ch4-1 Oscillation Characteristics of Lasers



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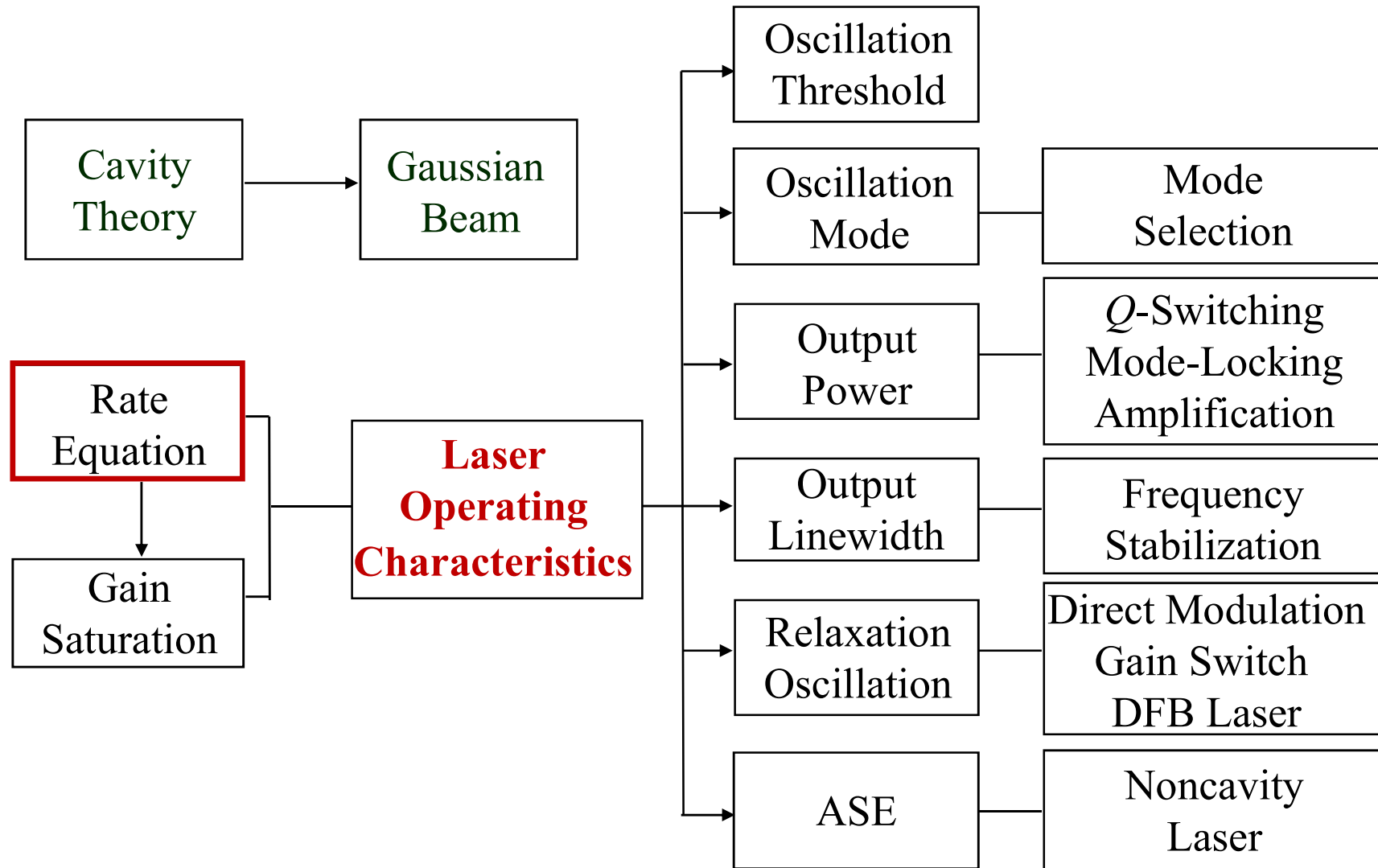
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Brief Introduction of Chapter 4-6



Chapter 4: Oscillation Characteristics

❖ Introduction

- Continuous wave (CW) lasers
- Pulse lasers

❖ 4.1 Oscillation Threshold

- Threshold condition for population inversion Δn_t
- Threshold condition for gain coefficient g_t
- Threshold condition for pumping power P_{pt}
- Threshold condition for pumping energy E_{pt}

Introduction: classification by pumping

➤ CW laser: Stable

During pumping, stable carrier at each energy level and a consistent photon density inside the cavity have been achieved., that is $dn/dt=0$; $dN_l/dt=0$

Rate equation $\longrightarrow$ Algebraic equation (代数方程)

➤ Pulsed laser: Nonstable

With a short pumping time, there will be significant variation of carriers at each energy level and photon density inside the cavity. Further classification is made based on the pumping time t_0 and lifetime of upper energy level τ_2

- ✓ Short-pulse laser ($t_0 < \tau_2$): pumping stops before reaching stable state.

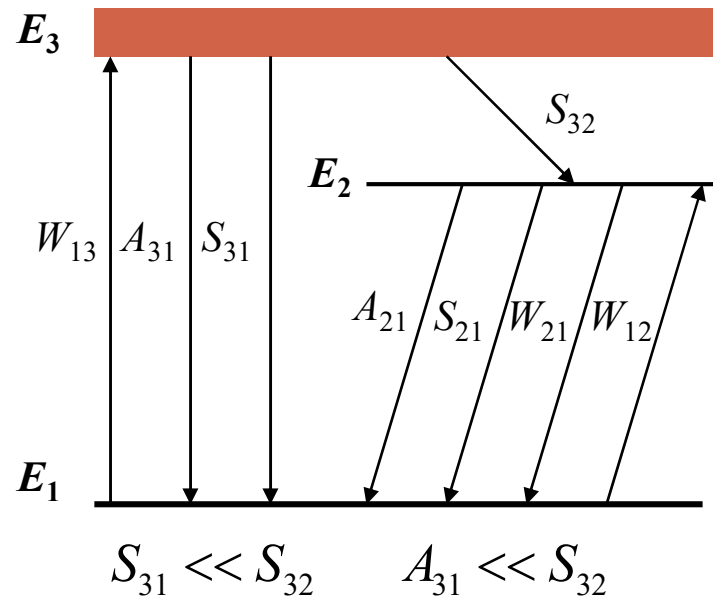
Nonstable; numerical solution; weak signal perturbation or other approximation methods.

- ✓ Long-pulse laser ($t_0 \gg \tau_2$): long pumping, approach stable state.

Can be treated as CW laser.

Theoretically, no strict boundaries between pulsed and CW lasers

❖ Rate equation for three-level system (ruby laser)



$$\begin{cases} \frac{dn_3}{dt} = n_1 W_{13} - n_3 (S_{32} + A_{31}) \\ \frac{dn_2}{dt} = n_1 W_{12} - n_2 W_{21} - n_2 (A_{21} + S_{21}) + n_3 S_{32} \\ n_1 + n_2 + n_3 = n \end{cases}$$

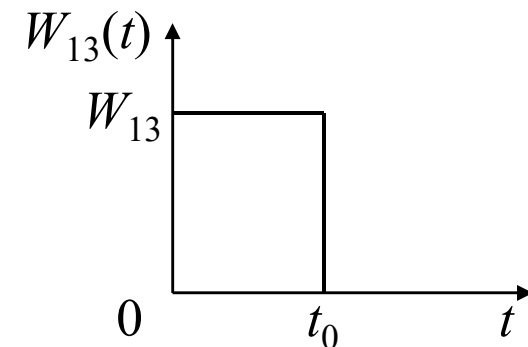
Consider the
cavity mode l

$$\frac{dN_l}{dt} = n_2 W_{21} - n_1 W_{12} - \frac{N_l}{\tau_{Rl}}$$

❖ Optical pumping of the three-level system

Rectangular
pulse excitation

$$\begin{cases} W_{13}(t) = W_{13} & 0 < t < t_0 \\ W_{13}(t) = 0 & t > t_0 \end{cases}$$



$$\frac{dn_3}{dt} = n_1 W_{13} - n_3 (S_{32} + A_{31}) \xrightarrow[n_3 \approx 0]{S_{32} \gg W_{13}} n_1 W_{13} = n_3 (S_{32} + A_{31})$$

$$\frac{dn_2}{dt} = - \left(n_2 - \frac{f_2}{f_1} n_1 \right) \sigma_{21}(\nu, \nu_0) \nu N_l - n_2 (S_{21} + A_{21}) + n_3 S_{32}$$

$$n_1 + n_2 + n_3 = n \rightarrow n_1 \approx n - n_2$$

Pump $\rightarrow$ Weak STE near threshold

$$\eta_1 = S_{32} / (S_{32} + A_{31}) \quad \text{Quantum efficiency of nonradiative recombination}$$

$$\eta_2 = A_{21} / (A_{21} + S_{21}) \quad \text{Quantum efficiency of fluorescence}$$

$$n_1 W_{13} = n_3 (S_{32} + A_{31}) \Rightarrow n_1 W_{13} = n_3 S_{32} / \eta_1$$

$$n_2 (S_{21} + A_{21}) \Rightarrow n_2 (S_{21} + A_{21}) = n_2 A_{21} / \eta_2$$

$$\frac{dn_2}{dt} = n_1 W_{13} \eta_1 - n_2 A_{21} / \eta_2 = (n - n_2) W_{13} \eta_1 - n_2 A_{21} / \eta_2$$

$$\frac{dn_2(t)}{dt} = \left[n - n_2(t) \right] W_{13}(t) \eta_1 - n_2(t) A_{21} / \eta_2$$

Solution

When $0 < t \leq t_0$, $W_{13}(t) = W_{13}$ $n_2(t) = \frac{\eta_1 W_{13} n}{A_{21} / \eta_2 + \eta_1 W_{13}} \left[1 - e^{-(A_{21} / \eta_2 + \eta_1 W_{13}) t} \right]$

Discussions

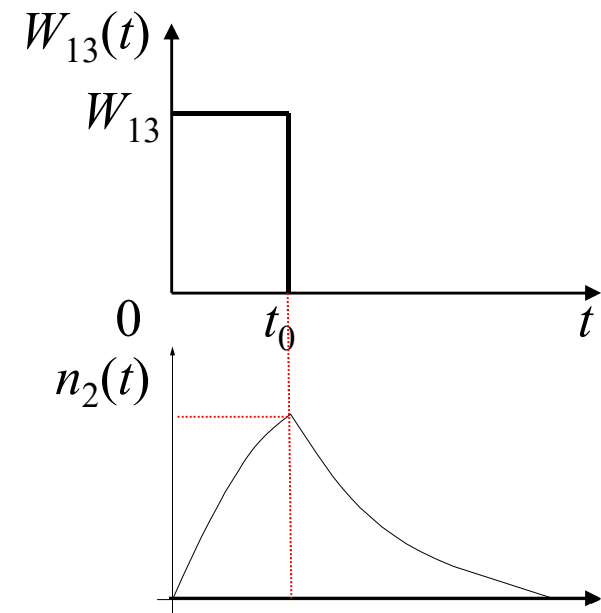
1. Two processes

- $0 < t < t_0$ During pumping $n_2 \uparrow$
- $t > t_0$ Pumping stopped $n_2 \downarrow$

$$W_{13} = 0 \Rightarrow n_3 S_{32} = 0 \Rightarrow \frac{dn_2}{dt} = -n_2 (A_{21} + S_{21})$$

2. For $t > t_0$ (n_2 reaches its maximum at $t = t_0$)

$$n_2(t) = n_2(t_0) e^{-\frac{A_{21}}{\eta_2}(t-t_0)}$$



3. $t_0 < \tau_2$ (short-pulse pumping) very short time, ignore SP

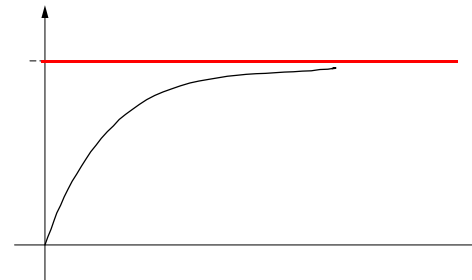
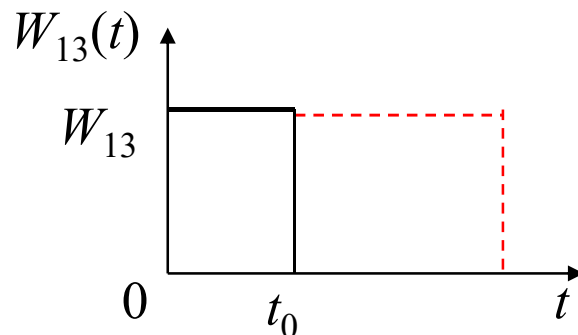
$$\Rightarrow \frac{dn_2(t)}{dt} = [n - n_2(t)]W_{13}\eta_1 \Rightarrow n_2(t) = n[1 - \exp(-\eta_1 W_{13}t)]$$

During pumping, $n_2(t)$ continuously increase: **nonstable**

4. $t_0 \gg \tau_2$ (long-pulse pumping) long enough time $\tau_2 = 1/(A_{21} + S_{21})$

$$n_2(t) \approx \frac{\eta_1 W_{13} n}{A_{21}/\eta_2 + \eta_1 W_{13}} \quad n_1(t) \approx n - n_2(t) \approx \frac{A_{21} n}{A_{21} + \eta_1 \eta_2 W_{13}}$$

n_2 continuously increases and reaches stable; n_1 reaches stable as well.



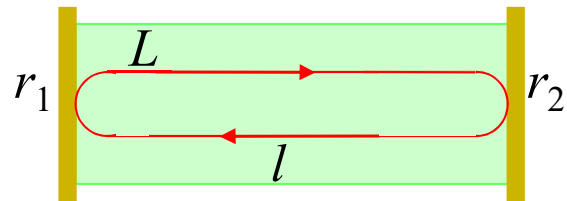
4.1 Oscillation Threshold

I. Population inversion Δn_t (threshold condition)

Requirements of self-oscillation: 1) $\Delta n > 0$; 2) $g > a$

Deduction of Δn_t : 1) Intensity change (光强变化); **2) Rate equation**

(1) Intensity change $\rightarrow \Delta n_t$ ($L=l$) L : cavity length; l : length of the gain medium



$$I_1 = I_0 e^{(g^0 - \alpha')l} r_2 e^{(g^0 - \alpha')l} r_1 = r_1 r_2 I_0 e^{2(g^0 - \alpha')l} \geq I_0$$

$$\alpha = \alpha_T + \alpha_d + \alpha_{s_1} + \alpha_{s_2} + \alpha_i \quad \underbrace{\hspace{10em}}_{\alpha'} \quad r_1 r_2 e^{2(g^0 - \alpha')l} \geq 1 \quad \Rightarrow \quad g^0 = \alpha' - \frac{1}{2l} \ln r_1 r_2 = g_t = \delta/l$$

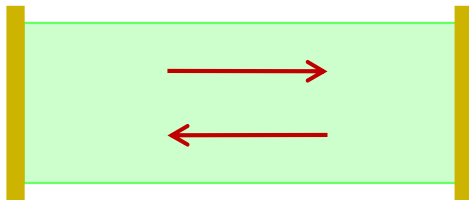
$$I_1 = I_0 e^{2(g^0 l - \delta)} \geq I_0 \quad \Rightarrow \quad g_t = \delta/l \quad \Rightarrow \quad \Delta n_t = \frac{g_t}{\sigma_{21}(\nu, \nu_0)} = \frac{\delta}{\sigma_{21}(\nu, \nu_0) l}$$

$$\delta = \alpha l$$

- conditions
- Cavity fully filled with gain material
 - Weak signal gain coefficient g^0 is a constant

(2) Rate equation $\rightarrow \Delta n_t (L=l)$

Weak signal: $\Delta n = \Delta n_t$ at $\frac{dN_l}{dt} = 0$



$$\frac{dN_l}{dt} = \left(n_2 - \frac{f_2}{f_1} n_1 \right) \sigma_{21}(\nu, \nu_0) \nu N_l - \frac{N_l}{\tau_{Rl}}$$

$$\frac{dN_l}{dt} = 0 \Rightarrow \Delta n_t \sigma_{21}(\nu, \nu_0) \frac{c}{\eta} = \frac{1}{\tau_{Rl}}$$



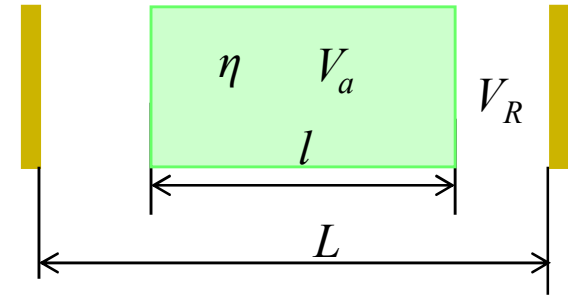
$$\because \tau_{Rl} = \frac{L'}{\delta c} = \frac{\eta l}{\delta c}$$

$$\Delta n_t \sigma_{21}(\nu, \nu_0) = \frac{\delta}{l} = g_t$$

$$\Delta n_t = \frac{g_t}{\sigma_{21}(\nu, \nu_0)} = \frac{\delta}{\sigma_{21}(\nu, \nu_0) l}$$

(2) Rate equation $\rightarrow \Delta n_t (L > l)$

$$\frac{dN_l}{dt} = \left(n_2 - \frac{f_2}{f_1} n_1 \right) \sigma_{21}(\nu, \nu_0) \nu N_l - \frac{N_l}{\tau_{Rl}}$$



$$V_a = Al \quad V_R = A(L-l) \quad \downarrow \text{Assume cross-section } A \text{ is constant inside the cavity}$$

Correction

$$\frac{d[N_l Al + N'_l A(L-l)]}{dt} = \left(n_2 - \frac{f_2}{f_1} n_1 \right) \sigma_{21}(\nu, \nu_0) \frac{c}{\eta} N_l Al - \frac{N_l Al + N'_l A(L-l)}{\tau_{Rl}}$$

$$\frac{L'A}{\eta} \left(\frac{dN_l}{dt} \right) \quad N_l \frac{c}{\eta} = N'_l c \quad \downarrow \quad \tau_{Rl} = \frac{L'}{\delta c} \quad L' = \eta l + L - l \quad \frac{L'A}{\eta} N_l$$

$$\frac{dN_l}{dt} = \left(n_2 - \frac{f_2}{f_1} n_1 \right) \sigma_{21}(\nu, \nu_0) c N_l \frac{l}{L'} - N_l \frac{\delta c}{L'}$$

$$\Delta n^0$$

$$dN_l/dt = 0$$

$$\Delta n^0 \geq \Delta n_t = \frac{\delta}{\sigma_{21}(\nu, \nu_0) l}$$

$$\Delta n^0 \geq \Delta n_t = \frac{\delta}{\sigma_{21}(\nu, \nu_0)l}$$

$$\sigma_{21}(\nu, \nu_0) = \frac{A_{21}\nu^2}{8\pi\nu_0^2} \tilde{g}(\nu, \nu_0)$$

➤ Different modes ν have different σ_{21} and therefore different Δn_t

➤ Minimum Δn_t at central frequency ν_0 $\Delta n_t(\nu_0) = \frac{\delta}{\sigma_{21}(\nu_0, \nu_0)l} = \frac{\delta}{\sigma_{21}l}$

➤ Threshold for population inversion at central frequency ν_0

$$\Delta n_t(\nu_0) = \Delta n_t = \frac{\delta}{\sigma_{21}l} \quad g = \Delta n \sigma_{21}(\nu, \nu_0) \quad \longrightarrow \quad g^0 \geq g_t = \Delta n_t \sigma_{21} = \frac{\delta}{l}$$

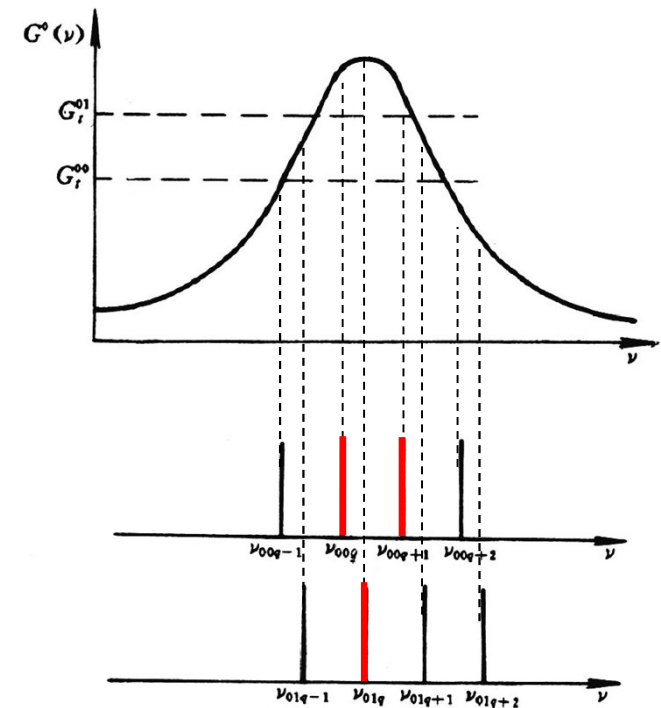
Threshold gain coefficient g_t is determined only by single-loop loss δ ;
A constant δ leads to a constant g_t .

II. Gain coefficient g_t (threshold condition)

$$g^0 \geq g_t = \frac{\delta}{l} = \Delta n_t \sigma_{21} \begin{cases} \Delta n_t = \frac{\delta}{\sigma_{21} l} = \frac{\nu^2 A_{21}}{4\pi^2 \nu_0^2 \Delta \nu_H} & \text{homogeneous broadening} \\ \Delta n_t = \frac{\delta}{\sigma_{21} l} = \frac{\sqrt{\ln 2} \nu^2 A_{21}}{4\pi^{3/2} \nu_0^2 \Delta \nu_D} & \text{inhomogeneous broadening} \end{cases}$$

Discussions

- Different modes $\nu \rightarrow$ different $\sigma_{21}(\nu, \nu_0) \rightarrow$ different Δn_t , so $\Delta n_t(\nu)$
- Different longitudinal modes have the same g_t
- Different transverse modes have different diffraction losses $\rightarrow$ different $g_t \rightarrow$ the threshold gain coefficient of high order transverse mode is higher than the fundamental mode $g_t^{01} > g_t^{00}$



III. Pumping power P_{pt} (threshold condition) for CW or long-pulse lasers (P_{pt} , $t_0 \gg \tau_2$)

1) Four-Level System (assume homogenous pumping)

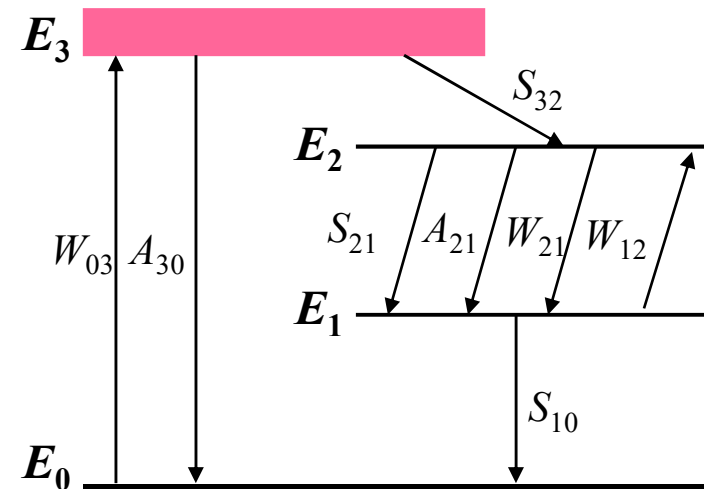
$$\because S_{10} \gg 0 \quad n_1 \approx 0 \quad \therefore \Delta n \approx n_2$$

$$\Rightarrow n_{2t} \approx \Delta n_t = \frac{\delta}{\sigma_{21} l} \quad \text{at central frequency}$$

When $n_2 = n_{2t}$ (treated as stable state)

For unit time and unit volume

- Transited carriers $E_2 \rightarrow E_1$ is n_2/τ_2 or $n_2/\tau_s \eta_2$, to maintain $n_2 = n_{2t}$
- Need carriers supply from $E_3 \rightarrow E_2$ as n_2/τ_2 or $n_2/\tau_s \eta_2$
- So via pumping (absorption) from $E_0 \rightarrow E_3$ is $n_2/\tau_2 \eta_1$ or $n_2/\tau_s \eta_1 \eta_2$



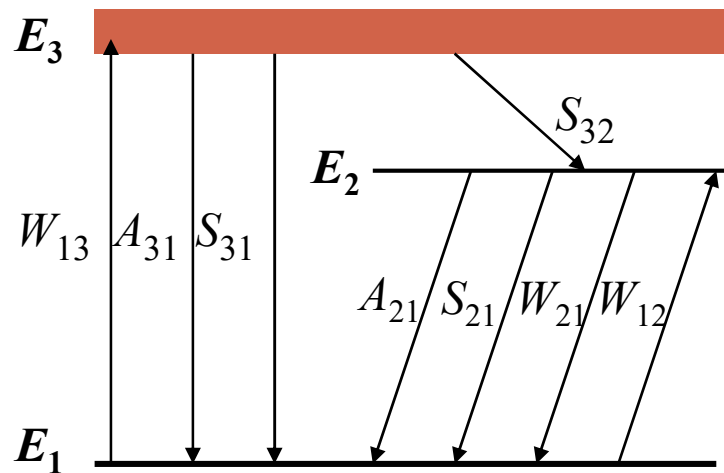
$$P_{\text{pt}} = \frac{h\nu_p \Delta n_t V}{\eta_1 \eta_2 \tau_s} = \frac{h\nu_p \Delta n_t V}{\eta_F \tau_s} = \frac{h\nu_p \delta V}{\eta_F \sigma_{21} \tau_s l}$$

$h\nu_p$ Photon energy of the pumping

$\eta_F = \eta_1 \eta_2$ Total quantum efficiency

2) Three-Level System (assume homogenous pumping)

- Similar analysis
- Only difference: the bottom level of the laser is ground state



$\Delta n = n_2 - n_1$ ignore degenerate of energy levels

$n_1 + n_2 \approx n$ because $n_3 \approx 0$

$$n_2 = \frac{1}{2}(n + \Delta n)$$

Assume threshold of population inversion at energy level E_2 is n_{2t}

$$n_{2t} = \frac{n + \Delta n_t}{2} \quad \Delta n_t \ll n \quad \Rightarrow$$

$$n_{2t} \approx \frac{n}{2} \quad \Rightarrow$$

$$P_{\text{pt}} = \frac{h\nu_p n V}{2\eta_F \tau_s}$$

IV. Pumping Energy E_{pt} (threshold condition) for short-pulse lasers (E_{pt} , $t_0 \ll \tau_2$)

For short-pulse pumping

➤ Ignore A_{21} and S_{21} , only consider pumping induced stimulated process

✓ In order to keep $n=1$, need to absorb photon from pumping is $1/\eta_1$

✓ In order to keep $n_2=n_{2t}$, need to absorb photon from pumping is n_{2t}/η_1

✓ To have lasing, the absorbed photon number from pumping per unit volume should be bigger than n_{2t}/η_1

Energy Level of Lasers	Long-pulsed or CW	Short-pulsed
Three-Level System	$P_{pt} = \frac{h\nu_p n V}{2\eta_1 \eta_2 \tau_s}$	$E_{pt} = \frac{h\nu_p n V}{2\eta_1}$
Four-Level System	$P_{pt} = \frac{h\nu_p \Delta n_t V}{\eta_1 \eta_2 \tau_s}$	$E_{pt} = \frac{h\nu_p \Delta n_t V}{\eta_1}$

For $t_0 \sim \tau_2$ (for case between long and short pulse), A_{21} and S_{21} , can NOT be ignored, NO analytical solution for pumping energy E_{pt} . If t_0 is given, we can numerically solve the problem.

Discussion

1) Threshold of four-level system is lower than three-level system

- ✓ Four-level system $n_1 \approx 0$

 - only need to pump Δn_t to achieve $g > a$ to have lasing

- ✓ Three-level system n_1 is ground state

 - at least need to pump $n/2$ and $n/2 \gg \Delta n_t$ to have lasing

2) To increase the pumping efficiency η_1

- ✓ Erbium-doped fiber amplifier (EDFA) use laser diode for pumping

3) P_{pt} and E_{pt} are related to the properties of gain material ($\eta_F, \sigma_{21}, \tau_s, \Delta\nu_F$)

$$P_{pt} = \frac{h\nu_p \delta V}{\eta_F \sigma_{21} \tau_s l}$$



$$\sigma_{21} = \frac{\nu^2 A_{21}}{4\pi^2 \nu_0^2 \Delta\nu_H} \quad \text{homogeneous broadening}$$

$$\sigma_{21} = \frac{\sqrt{\ln 2} \nu^2 A_{21}}{4\pi^{3/2} \nu_0^2 \Delta\nu_D} \quad \text{inhomogeneous broadening}$$

$$\eta_F \uparrow, \sigma_{21} \uparrow, \tau_s \uparrow \Rightarrow P_{pt}, E_{pt} \downarrow$$

$$\Delta\nu_F \downarrow \rightarrow \sigma_{21} \uparrow \rightarrow P_{pt}, E_{pt} \downarrow$$

Materials with narrow fluorescent linewidth is preferred for gain material.

Type of lasers	Ruby	Nd:Glass	Nd:YAG	He-Ne
$\Delta\nu_F$ (Hz)	3.3×10^{11}	7×10^{12}	1.95×10^{11}	1.5×10^9
τ_s (s)	3×10^{-3}	7×10^{-4}	2.3×10^{-4}	7×10^{-9}
Δn_t (cm ⁻³)	8.7×10^{17}	1.4×10^{18}	1.8×10^{16}	10^9
n_{2t} (cm ⁻³)	9.5×10^{18}	1.4×10^{18}	1.8×10^{16}	
η_F	0.7	0.4	1	
E_{pt}/V (J/cm ³)	5	0.95	4.9×10^{-3}	
P_{pt}/V (W/cm ³)	1600	1400	21	

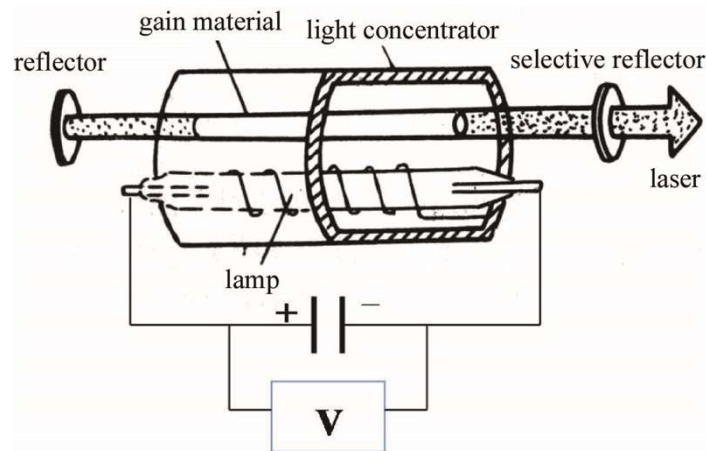
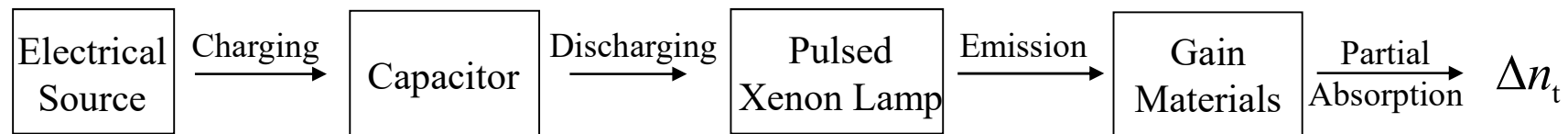
4) For four-level system $\Delta n_t \propto \delta$ $\delta \uparrow \rightarrow P_{pt}, E_{pt} \uparrow$

- ✓ Must make sure the quality of the optical component inside the cavity in order to decrease the optical loss.

5) Meaning of P_{pt} and E_{pt}

- ✓ The derived P_{pt} and E_{pt} : Effective pumping power and energy
- ✓ The laser's P_{pt} and E_{pt} : Electrical power and energy of input pumping light source

for solid-state lasers



- Solid-state lasers: optical pumping
- Semiconductor lasers: injection current (I_{th})
- Gas lasers: discharge current